The Nutraceutical PEA as a Glucocorticoid-Sparing Agent: A Novel PPARα-GR Cross-Talk Approach to Safer Anti-Inflammatory Treatment of Rheumatoid arthritis.

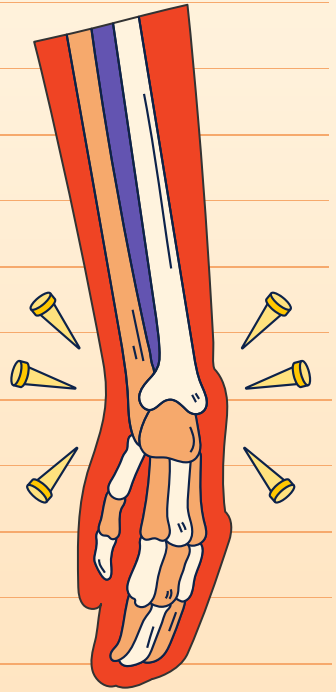
Supervised By:
Associate. Prof Ahmed abo-elfotoh Morad
Associate. Prof Ahmed Gafaar Gafaar
Associate. Prof Ahmed Saad Saad

By: T.A Riffat Ibrahim Riffat

Introduction

Rheumatoid arthritis (RA) is a debilitating chronic inflammatory autoimmune disease that affects approximately 1% of the global population.

Its hallmark is a symmetrical polyarthritis, predominantly affecting the small joints of the hands and feet.



Introduction

The pathophysiology involves a complex interplay of genetic and environmental factors that trigger a dysregulated immune response.

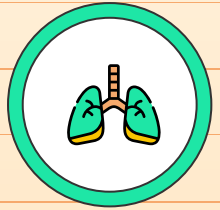
that leads to synovial inflammation (synovitis), proliferation of the synovial lining (pannus formation), and subsequent erosion of cartilage and bone.

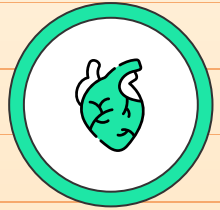




Introduction

Importantly, RA is a systemic condition, often presenting with extra-articular features such as fatigue, rheumatoid nodules, and an increased risk of cardiovascular and pulmonary diseases





The significant morbidity and socioeconomic burden associated with RA underscore the importance of early diagnosis and aggressive treatment to induce remission or low disease activity, thereby preserving joint integrity and patient function.

Treatment of R.A

Conventional Synthetic DMARDs (csDMARDs)	First-line disease-modifying therapy	Methotrexate
Antimalarials (csDMARD)	Mild RA; used in combination therapy	Hydroxychloroquine
Immunosuppressants (csDMARD)	Alternative to MTX; slows disease progression	Leflunomide

Treatment of R.A

TNF-α Inhibitors (Biologic DMARDs)	Target cytokines to reduce inflammation	Etanercept
IL-6 Inhibitors (Biologic DMARDs)	Reduce systemic inflammation	Tocilizumab
T-cell Co-stimulation Modulator (Biologic)	Blocks T-cell activation	Abatacept

Treatment of R.A

B-cell Depleting Agent (Biologic)	Reduces autoantibody production	Rituximab
JAK Inhibitors (Targeted Synthetic DMARDs)	Oral targeted therapy; reduces cytokine activity	Tofacitinib

Treatment of R.A

NSAIDs	Symptomatic relief of pain & inflammation (no effect on disease progression)	Ibuprofen
Corticosteroids	**Rapid control of acute inflammation; bridge therapy**	**Prednisolone**



The role of Glucocorticoids in RA

Glucocorticoids are potent anti-inflammatory and immunosuppressive agents that play several critical, but typically temporary, roles in RA management:

"Bridge" Therapy: This is their primary role. When a patient is started on a slow-acting DMARD (like methotrexate), which can take weeks or months to become fully effective, glucocorticoids provide rapid relief from pain, swelling, and stiffness. They "bridge" the patient until the DMARD starts working.



The role of Glucocorticoids in RA

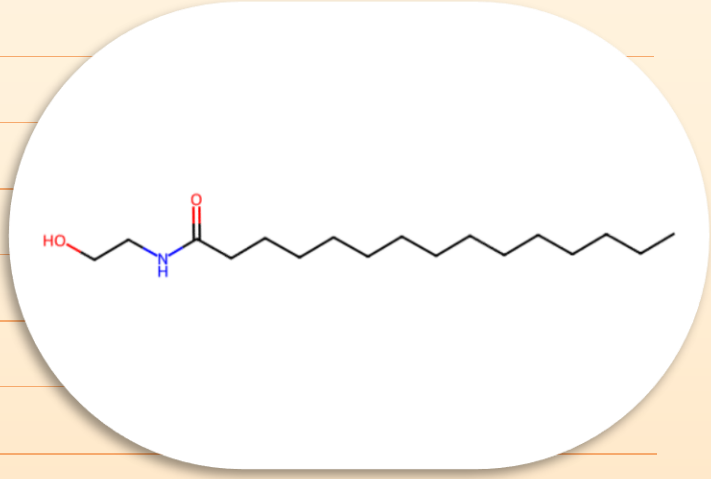
Long-term Low-Dose Therapy: A small subset of patients with severe, difficult-to-control disease may require a very low maintenance dose in addition to their other DMARDs to maintain stability, though this is avoided if possible.

Disease-Modifying Potential: At higher doses, glucocorticoids can genuinely slow the progression of joint damage seen on X-rays, a property they share with DMARDs.

Despite their powerful efficacy, glucocorticoids are used with extreme caution and at the lowest possible dose for the shortest possible duration due to a wide range of dose- and duration-dependent side effects such as metabolic and CVS complications.

What is PEA

- **N-Palmitoylethanolamine (PEA) is an endogenous lipid mediator, chemically composed of palmitic acid and ethanolamine, which is naturally produced within the human body and found in dietary sources such as egg yolks, soybeans, and peanuts.**



Wang Y, Duan X, Li Z, Pan Y, Deng J. Palmitoylethanolamide in the Treatment of Pain and Its Clinical Application Prospects. Drug Des Devel Ther. 2025;19:6897-6923.

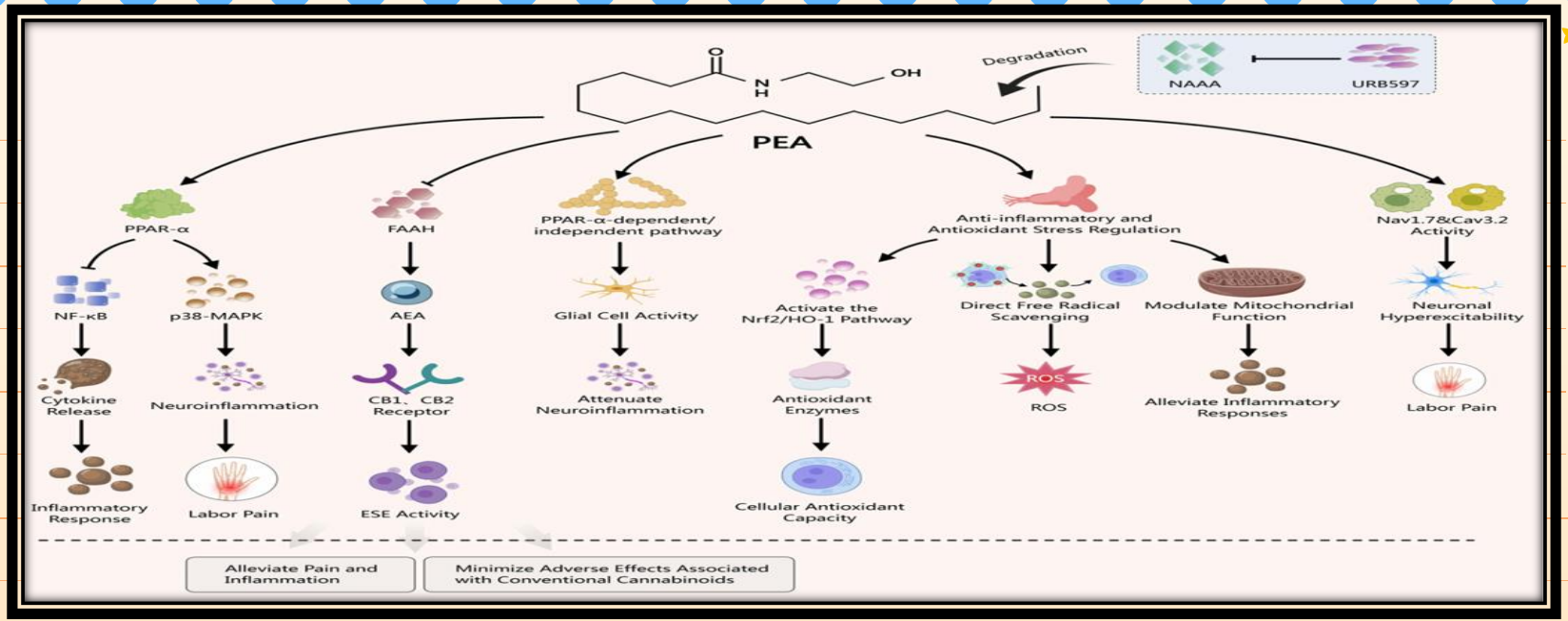
What is PEA

Due to its important neuroprotective, anti-inflammatory, and analgesic actions, PEA has emerged as a promising nutraceutical. It exhibits notable therapeutic effects alongside an extremely low incidence of adverse reactions, demonstrating a high safety profile even in special populations like the elderly and pregnant women.

This favorable safety and broad therapeutic window make PEA suitable not only as a standalone therapy but also as a highly appropriate adjunct medication. In combination treatments, it can reduce the required dosage of primary drugs such as NSAIDs and opioids, thereby mitigating the associated risk of their side effects.



Wang Y, Duan X, Li Z, Pan Y, Deng J. Palmitoylethanolamide in the Treatment of Pain and Its Clinical Application Prospects. Drug Des Devel Ther. 2025;19:6897-6923.



Wang Y, Duan X, Li Z, Pan Y, Deng J. Palmitoylethanolamide in the Treatment of Pain and Its Clinical Application Prospects. Drug Des Devel Ther. 2025;19:6897-6923.

PEA as an Anti-inflammatory

While PEA exerts its anti-inflammatory effects through multiple modes of action, its primary mechanism involves the activation of PPAR-α.

This activation regulates key signaling pathways, leading to the inhibition of both NF-κB and p38-MAPK, which collectively reduce pro-inflammatory factors and modulate neuroinflammation.

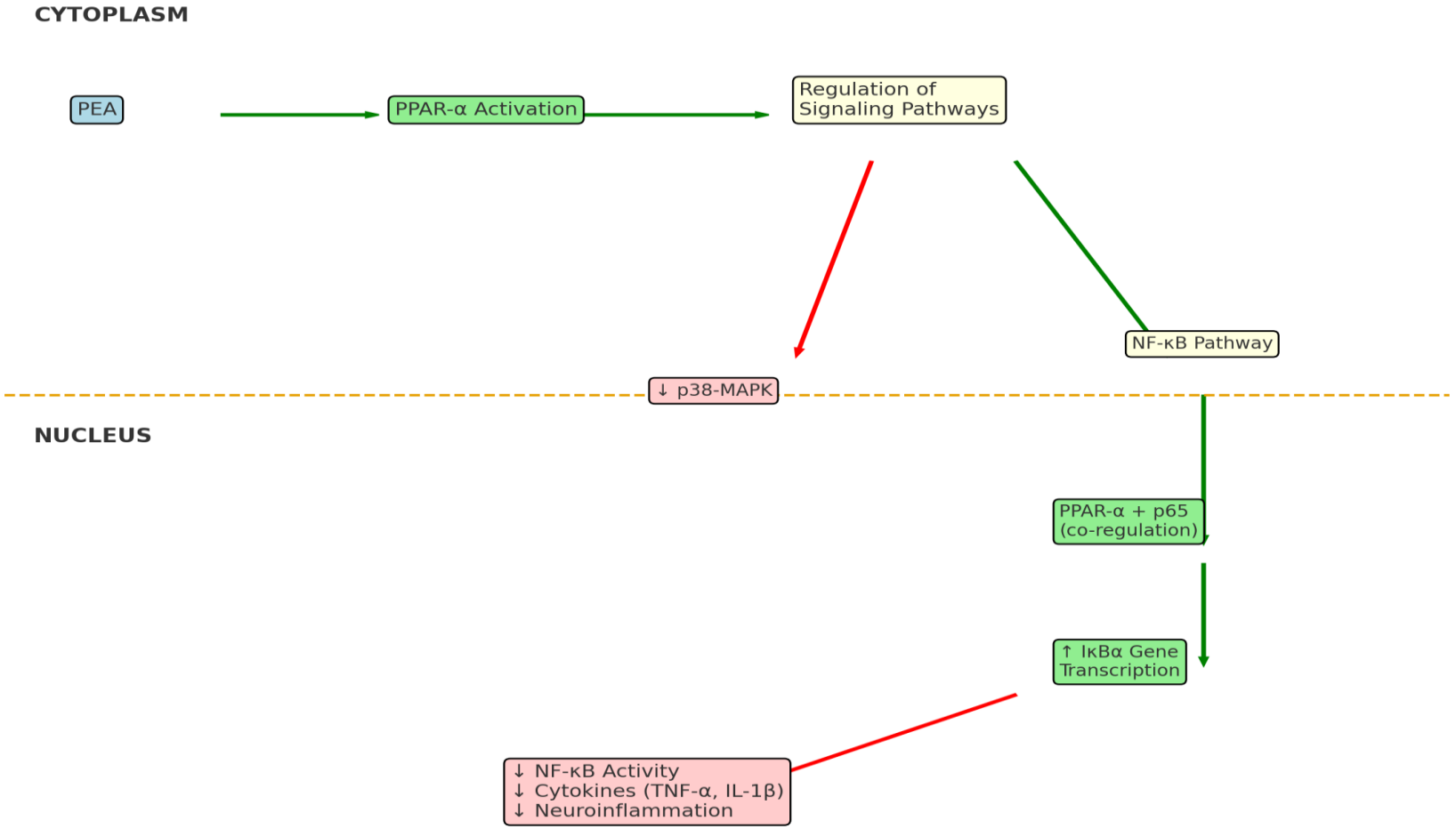
Wang Y, Duan X, Li Z, Pan Y, Deng J. Palmitoylethanolamide in the Treatment of Pain and Its Clinical Application Prospects. Drug Des Devel Ther. 2025;19:6897-6923.

PEA as an Anti-inflammatory

The specificity of the NF-κB inhibition occurs through a novel transcriptional mechanism independent of classical PPAR Response Elements (PPREs).

Instead, in a ligand-dependent manner, PPAR-α potentiates the ability of the NF-κB subunit p65 to activate the promoter for IκBα, the natural inhibitor of NF-κB. Thus, PPAR-α reduces inflammation by indirectly boosting IκBα expression, thereby suppressing NF-κB activity.

Wang Y, Duan X, Li Z, Pan Y, Deng J. Palmitoylethanolamide in the Treatment of Pain and Its Clinical Application Prospects. Drug Des Devel Ther. 2025;19:6897-6923.


CYTOPLASM
PEA
PPAR-α Activation
Regulation of Signaling Pathways
NF-κB Pathway
↓ p38-MAPK
NUCLEUS
PPAR-α + p65 (co-regulation)
↑ IκBα Gene Transcription
↓ NF-κB Activity
↓ Cytokines (TNF-α, IL-1β)
↓ Neuroinflammation

Cross-Talk with the Glucocorticoid Receptor (GR):

PPARα and GR converge at multiple levels to reinforce anti-inflammatory and metabolic regulation:

1. GR Regulation of PPARα Expression:
GR directly upregulates PPARα transcription in hepatocytes, aligning PPARα expression with the diurnal rhythm of circulating glucocorticoids and stress responses.

Bougarne, N., Weyers, B., Desmet, S. J., Deckers, J., Ray, D. W., Staels, B., & De Bosscher, K. (2018). Molecular actions of PPARα in lipid metabolism and inflammation. *Endocrine Reviews*, *39*(5), 760–802. https://doi.org/10.1210/er.2018-00064

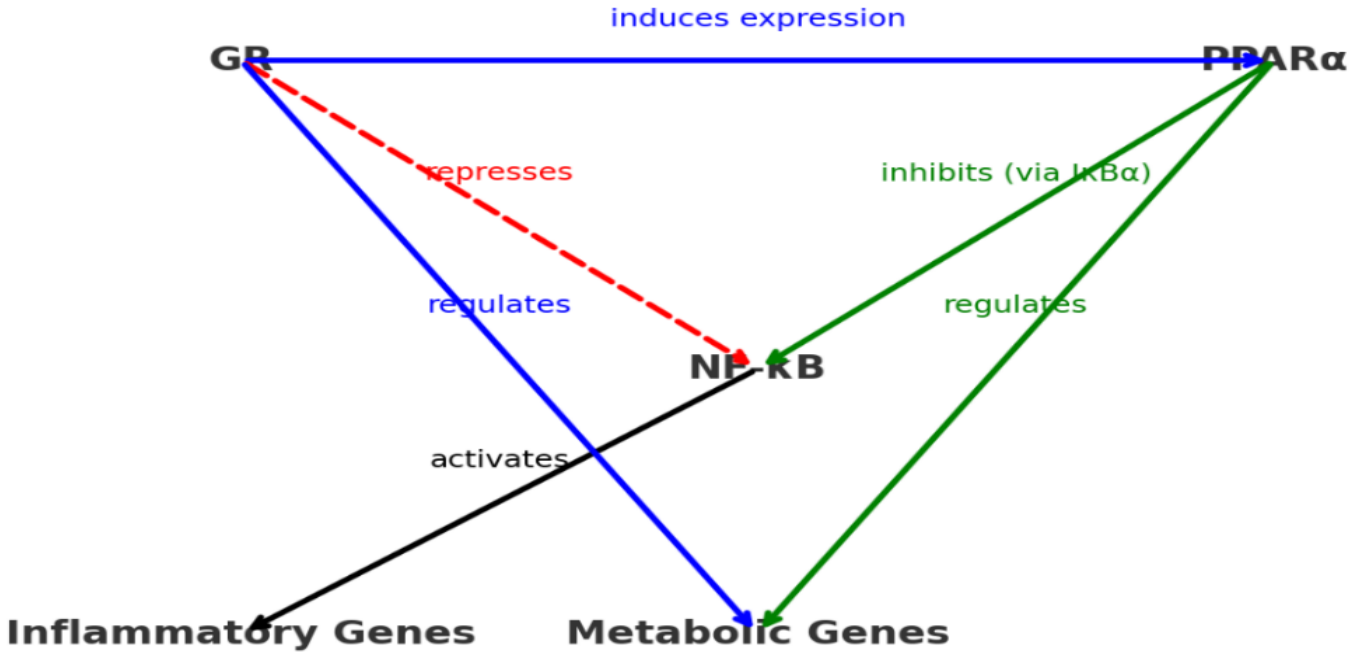
Cross-Talk with the Glucocorticoid Receptor (GR):

2. Synergistic Anti-Inflammatory Action: Both receptors suppress NF-κB signaling:

o GR interferes physically with p65 and recruits HDACs to inflammatory promoters.
o PPARα promotes IκBα expression and competes for limited coactivators.
When activated together, they exert synergistic repression of NF-κB target genes (e.g., MCP-1, IL-6), likely through direct protein-protein interactions.

Bougarne, N., Weyers, B., Desmet, S. J., Deckers, J., Ray, D. W., Staels, B., & De Bosscher, K. (2018). Molecular actions of PPARα in lipid metabolism and inflammation. *Endocrine Reviews*, *39*(5), 760–802. https://doi.org/10.1210/er.2018-00064

Simplified Cross-Talk Between PPARα and GR



Therapeutic Implications

Why This Is a Big Deal?

Understanding this partnership opens up exciting new possibilities for medicine, especially for treating chronic inflammatory diseases like asthma, rheumatoid arthritis, or inflammatory bowel disease.

- The Problem with "Crisis Manager" (GR) Alone: Using steroid drugs (which activate GR) is very effective but has severe side effects like weight gain, diabetes, and bone thinning when used long-term.

- The Potential Solution: Combining a lower, safer dose of a steroid (GR activator) with a PPARα-activating drug could be much more effective than either drug alone.

Rational

to use PEA to reduce the dose of a glucocorticoid the most appropriate and justifiable choice is <u>Prednisolone.</u>

Glucocorticoid	Rationale for Use	Rationale Against Use / Considerations
Prednisolone	- "Gold Standard" Oral GC - - Active Form (no liver conversion) - Extensive precedent in rat arthritis models - - Classic broad-spectrum anti-inflammatory GC (GR-p65 interference, HDAC recruitment)	None noted
Dexamethasone	- High potency & specificity (pure GR agonist) - Well-defined dosing with low doses	- Less clinically relevant for chronic RA - May overpower system, masking subtle synergistic effects of PEA

Rational

Justification:

Clinical Relevance: this thesis aims to solve a real-world clinical problem (GC side effects).

Experimental Reliability: Using the active form (prednisolone) eliminates potential variability in liver metabolism that you'd get with prednisone, leading to cleaner and more reproducible results.

Mechanistic Alignment: It effectively engages the GR-mediated anti-inflammatory pathways (like NF-κB repression) that are central to the proposed cross-talk with PPARα.

Study design

Aspect	Proposed Protocol
Animal Model	Adjuvant-Induced Arthritis (AIA) in rats
Experimental Groups	1Healthy controls (no arthritis) 2. Arthritic, untreated 3. Arthritic + effective dose of prednisone 4. Arthritic + sub-therapeutic dose of prednisone 5. Arthritic + PEA alone **6. Key Group: Arthritic + sub-therapeutic prednisone + PEA**
Treatment Duration	Begin treatment after disease onset (e.g., day 21-25 post-immunization) and continue for 3-4 weeks to assess chronic effects
PEA Formulation	Ultramicronized PEA (um-PEA). Ultramicronization enhances bioavailability and efficacy compared to non-micronized forms

Compound	Suggested Dose (I.P.)	Basis of the Suggestion & Evidence Gaps
Prednisolone	2-3 mg/kg (Effective Dose)	Based on a 2025 study with 10 mg/kg (oral) fully effective dose. IP dose adjusted for oral bioavailability (~70%). No direct IP RA dose found.
Prednisolone	0.5-1 mg/kg (Low/Sub-therapeutic Dose)	Based on same 2025 study where 5 mg/kg (oral) half-dose effective in combination therapy. Supports dose-reduction goal.
PEA	10-30 mg/kg	No specific IP RA dose found. Conservative estimate from typical experimental dosing. **Pilot study recommended.**

Morvaridi A, Abtahi Froushani SM, Farshid AA. Benefits of combining piperine with prednisolone in an experimental model of rheumatoid arthritis. Vet Res Forum. 2025;16(2):117-124. doi: 10.30466/vrf.2024.2031860.4316. Epub 2025 Feb 15. PMID: 40099241; PMCID: PMC11910975.

Balaji S, Woodward TJ, Richter E, Chang A, Otiz R, Kulkarni PP, et al. Palmitoylethanolamide causes dose-dependent changes in brain function and the lipidome. Front Neurosci. 2024;18:1506352.

Key Outcome Measures and Biomarkers

Clinical Disease Scoring: Monitor and score paw swelling, erythema, and joint deformation several times weekly using a standardized arthritis index.

Histopathological Analysis: After sacrifice, process joint tissues for histology. Stains like Hematoxylin & Eosin (H&E) and Safranin O can evaluate synovitis, pannus formation, cartilage erosion, and bone destruction

Molecular Biomarkers: Analyze serum, synovial fluid, or joint tissue homogenates for:

Pro-inflammatory Cytokines: TNF-α, IL-1β, IL-6, and PGE_2 via ELISA, which are known to be suppressed by PEA .

Molecular Analysis to Investigate Synergy

Protein Expression Analysis: Use Western Blot or Immunohistochemistry on joint tissue lysates to measure:

NF-κB Pathway: Phosphorylation of IκBα and nuclear translocation of p65.

Anti-inflammatory Target: Total IκBα protein levels, which PPARα is known to upregulate

Gene Expression Analysis: RT-qPCR on joint tissue to examine the expression of genes downstream of NF-κB (e.g., Tnf-α, Il-6) and GR.

Receptor Localization: immunohistochemistry to visualize the cellular co-localization of PPARα and GR in the synovium of treated animals, which could support a direct protein-protein interaction.

Kits for the RA Animal Model

Induction Reagents: Collagen (e.g., Chicken Type II Collagen) and an Immunoadjuvant (e.g., Complete Freund's Adjuvant, CFA) to induce arthritis.

Clinical Scoring Supplies: Basic calipers for measuring paw thickness and swelling.

Kits for Mechanism and Efficacy Analysis

Research Area	Required Kits & Reagents	Key Measurements
Molecular Mechanism	PPARα Reporter Assay Kit	PPARα activation by PEA
	ELISA Kits for IkBα & p65	NF-κBpathway protein levels (phospho-IkBα, nuclear p65)
Inflammation Assessment	ELISA Kits for IL-6, TNF-α, MCP-1	Inflammatory cytokine levels in serum/joint tissue

General Laboratory Supplies

Test Compounds: Prednisolone and Palmitoylethanolamide (PEA), both prepared in suitable vehicles for intraperitoneal injection.

Animal Injection Supplies: Appropriate syringes and needles.

Sample Collection Tubes: For collecting and processing blood (serum/plasma) and tissue samples.